

Orthocenter Bonanza

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Orthocenter is one of the favorite points in Olympiad Geometry. Here, we list some common configurations which they are associated with, with loads of problems in the end (you might have seen many, but this time try to associate them with the lemmas provided, and see how the tables turn!). Some (many) of these might be known to the reader, or might be trivial to prove, but even the most trivial properties are very useful in finding elegant solutions! There are many (really many) properties of the orthocenter, which can be found in the references section, but I decided to include the ones that I encountered the most while solving Olympiad Geometry problems. Enjoy!

Also, I would like to thank the user *Synthetic_Potato* [9] for his helpful suggestions, contributions and ideas.

‡ A Salad of Configurations

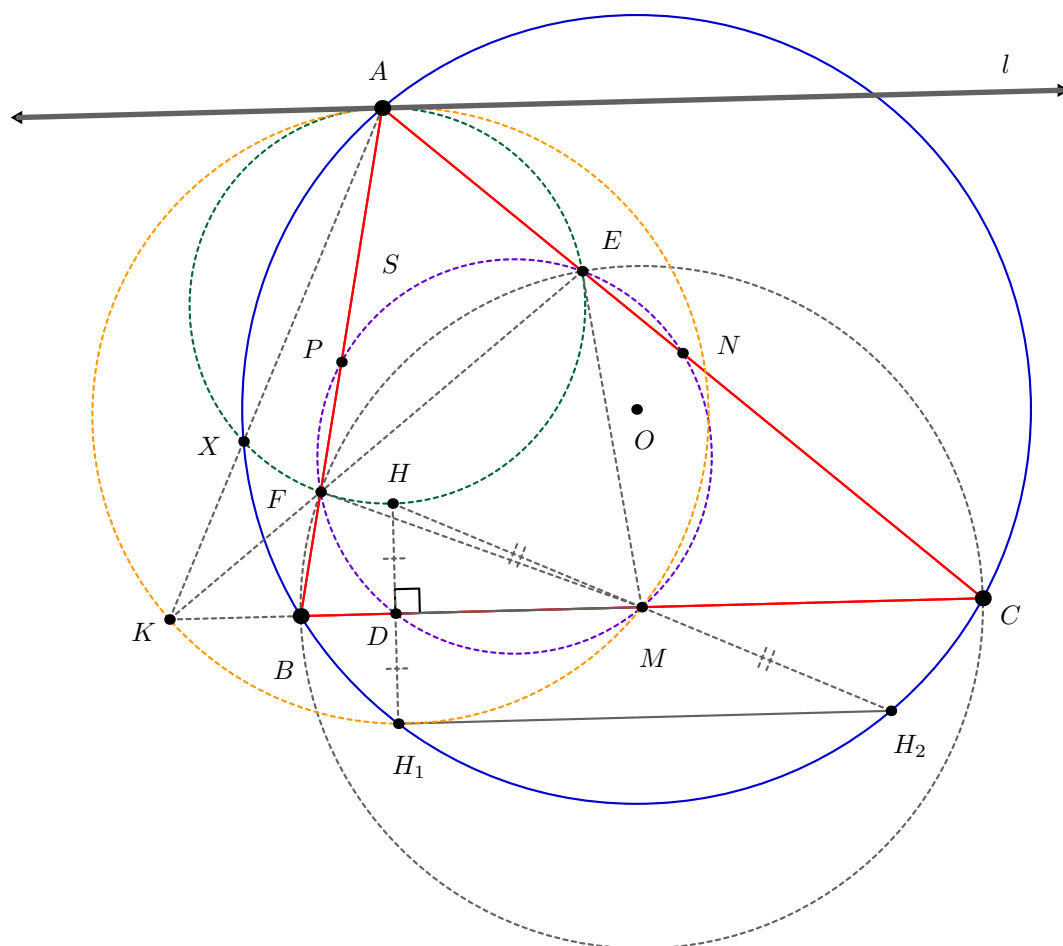


Figure 1: The Big Picture

There's a lot going on in the figure, and we will prove these results in a series of lemmas.

Firstly, we define each point (and some notations):

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- A, B, C are the standard vertices of the main triangle, $\angle CAB = \alpha$, $\angle ABC = \beta$ and $\angle BCA = \gamma$.
- D, E, F are the feet of the A, B, C altitudes respectively and H is the orthocenter, while M, N, P are the midpoints of BC, CA, AB respectively.
- $\odot(k)$ represents the circumcircle of k , $z \cap w$ represents the intersection of z and w . Note that all the angles used are directed and considered modulo 180° , unless otherwise specified.
- $H_1 = HD \cap \odot(ABC) \neq A$, $K = EF \cap BC$, $H_2 = HM \cap \odot(ABC)$ on the arc BC not containing A .
- B', C' represents the antipodes of B, C with respect to $\odot(ABC)$.
- l is a line parallel to BC through A and S is midpoint of AH .
- O is the center of $\odot(ABC)$ with radius R . Also, R_{LMN} represents the radius of $\odot(LMN)$.

It is worth noting the interesting (and useful) duality:

If H is the orthocenter of $\triangle ABC$, then A is also the orthocenter of $\triangle HBC$, and similar results for B and C .

Lemma 1.1: The reflection of H over BC lies on $\odot(ABC)$.

Proof : This is just angle chasing;

$\angle H_1BC = \angle H_1AC = \angle DAC = 90^\circ + \angle DCA = 90^\circ + \angle BCE = \angle CBE = \angle DBH$ and so H_1 is the reflection of H over BC , which, by definition, lies on $\odot(ABC)$. Hence we are done.

[I am again reminding the reader that the angles are directed and consider modulo 180° .]

Lemma 1.2 The reflection of H over M lies on $\odot(ABC)$, which is the point H_2 , and further AH_2 is a diameter of $\odot(ABC)$.

Proof : This is again simply angle chasing, but has a rather nice proof using complex numbers (vectors also work) which I present: (small letter represent complex numbers)

Set the circumcenter of ABC to be the origin. Let the reflection be J . Then clearly, $BHCJ$ is a parallelogram.

Thus, $j = b + c - h = -a$ and so $|j| = |-a| = |a| = R$ along with $|AJ| = |a - j| = 2|a| = 2R$ and hence both the results follow.

Lemma 1.3: $|DM| = R \sin(\beta - \gamma)$.

Proof : Note that $\angle DAO = \alpha - (\angle BAD + \angle CAO) = 180^\circ - \beta - \gamma - 2(90^\circ - \beta) = \beta - \gamma$ and by lemma 2, A, O, H_2 are collinear so we get $\angle H_1AH_2 = \beta - \gamma$.

Hence, by applying the sine rule to $\triangle H_1AH_2$, we get $|H_1H_2| = 2R \sin(\beta - \gamma)$.

In $\triangle HH_1H_2$, DM is the midline by lemma 1.1 and 1.2 and so $|DM| = R \sin(\beta - \gamma)$, as desired.

Lemma 1.4: $R_{BHC} = R_{CHA} = R_{AHB} = R$

Proof : By the sine rule applied to $\triangle BHC$, we get

$$2R_{BHC} = |BC| \csc(180^\circ - \alpha) = |BC| \csc \alpha = 2R, \text{ giving the result.}$$

The others follows similarly.

Lemma 1.5: $AH \parallel OM$ and $|AH| = 2|OM|$

Proof : It is clear that they are parallel, since the both are perpendicular to BC .

By right $\triangle BOM$, we get $|OM| = |BO| \cos \alpha = R \cos \alpha$ and by the sine rule applied to $\triangle AHB$, we get $|AH| = 2R \cos \alpha$, since $R_{AHB} = R$ by lemma 1.4.

Hence, $|AH| = 2|OM|$.

Or, we can directly use complex numbers for the second part;

$|AH| = |a - (a + b + c)| = |b + c| = 2|OM|$, if we set O to be the origin of the complex plane.

Lemma 1.6: (The Nine Point Circle) The midpoints of AH, BH, CH and the points D, E, F, M, N, P are all concyclic. The center of this circle is the midpoint of O and H , with radius $= 0.5R$. Also, $|SM| = R$.

Proof : One elegant proof is by considering a homothety centered at H with ratio 0.5, and then using lemma 1.1 and 1.2. But here we provide a different proof using complex numbers:

Consider the point N_9 to be the midpoint of OH , and set O to be the origin of the complex plane. We will show that N_9 is equidistant from all the nine points (in fact, by symmetry, showing it for 3 suffices)

Then

$$\begin{aligned}
|N_9M| &= |n_9 - m| = \left| \frac{a+b+c}{2} - \frac{1}{2}(b+c) \right| = \left| \frac{a}{2} \right| = \frac{R}{2} \\
|N_9S| &= |n_9 - s| = \left| \frac{a+b+c}{2} - \frac{1}{2}(a+b+c+a) \right| = \left| \frac{-a}{2} \right| = \frac{R}{2} \\
|N_9D| &= |n_9 - d| = \left| \frac{a+b+c}{2} - \frac{1}{2} \left(a+b+c - \frac{bc}{a} \right) \right| = \left| \frac{bc}{2a} \right| = \frac{R}{2}
\end{aligned}$$

and so the result follows. ($|SM| = R$ can also be easily shown in a similar fashion.)

Another really nice proof for the midpoints is to note that since $(K, D; B, C) = -1$, hence $KF \cdot KE = {}^{\odot(BFEC)} KB \cdot KC = KD \cdot KM$ and so M lies on $\odot(DEF)$. Similarly, the other midpoints lie on this circle.

Lemma 1.7: The points B, F, E, C are concyclic with center M .

Proof : Trivial.

Lemma 1.8: The points A, H, F, E are concyclic with center S .

Proof : Trivial.

Lemma 1.9: Let Y be a point on AD . Then $AD \cdot YD = BD \cdot DC$ if and only if Y is the orthocenter of $\triangle ABC$.

Proof : Reflect Y over BC . We know (by lemma 1.1) that Y lies on $\odot(ABC)$ if and only if $Y = H$. The rest is just Power of a Point on quadrilateral $ABY'C$, where Y' is the reflection of Y .

Lemma 1.10: $MH \perp AX$

Proof : This follows by Brocard's theorem on cyclic quadrilateral $FECB$ which has the center M (lemma 1.7). (A different proof without Brocard's can be found in the following section.)

Lemma 1.11: (The Three Tangents Lemma) MF, ME and l are all tangent to $\odot(AFHE)$.

Proof : Note that by right $\triangle BFC$, we get $|FM| = |MC|$. Also by right $\triangle EBC$, we get $|EM| = |MC|$. Hence, $|FM| = |ME|$ and so $\triangle MFE$ is isosceles.

Now, we know that $FDME$ are concyclic by lemma 1.6, and thus $\angle EMF = \angle EDF = 180^\circ - 2\alpha$, and so $\angle MFE = \angle FEM = \alpha = \angle FAE$, and so MF, ME are tangent to $\odot(AFE)$.

Now, S is the center of $\odot(AFE)$ (by lemma 1.8) and since A, S, D are collinear, hence $DA \perp l \Rightarrow SA \perp l$, and thus l is tangent to $\odot(AFE)$, as desired.

Another proof for the first two tangents uses the fact that M is the antipode of S with respect to the nine point circle, which gives us $\angle SPM = \angle SEM = 90^\circ$, and since S is the center of $\odot(AFHE)$, we are done.

Lemma 1.12: The points K, H_1, M, A are concyclic.

Proof : It suffices to show $DK \cdot DM = DA \cdot DH_1 = DB \cdot DC$, by Power of a point This can now be done by some trigonometric computations, as $|DM| = R \sin(\beta - \gamma)$ by lemma 1.3. Now, note that

$$|DM| = R \sin(\beta - \gamma) = \frac{c \sin(\beta - \gamma)}{2 \sin \gamma} = \frac{|DB| \sin(\beta - \gamma)}{2 \cos \beta \cdot \sin \gamma}$$

And finally,

$$|DK| = |DE| \cdot \frac{\sin(\pi - 2\beta)}{\sin(\beta - \gamma)} = \left(|DC| \cdot \frac{\sin \gamma}{\sin \beta} \right) \left(\frac{\sin(2\beta)}{\sin(\beta - \gamma)} \right) = \frac{|DC| \cos \beta \cdot \sin \gamma}{\sin(\beta - \gamma)}$$

where the first equality follows from applying sine rule to $\triangle DKE$ and the second one by $\triangle DEC$. Multiplying these two, we get the desired result. [The reader is requested to fill in the easy angle chasing details]

A synthetic proof goes something like this:

It suffices to show $DK \cdot DM = DA \cdot DH_1 = DA \cdot DH$, as $DH = DH_1$ by lemma 1.1.

Now, by Brocard's theorem (or say, lemma 1.10, or even lemma 2.1) we get that H is also the orthocenter of $\triangle AMK$. Thus, by lemma 1.9 on $\triangle AMK$, we get $AD \cdot DH = KD \cdot DM$, completing the proof.

Note that this also gives us that AD is the radical axis of $\odot(ABC)$ and $\odot(MAK)$.

Lemma 1.13 (Euler line) The Orthocenter, Centroid and the Circumcenter (and in fact the nine-point center too) are collinear. Further, $2|OG| = |GH|$, where G is the centroid.

Proof : Consider a homothety centered at G with ratio -2 . This sends the medial triangle to $\triangle ABC$. Now, it is easy to show that O is the orthocenter of the medial triangle, and so the image of O is H , giving the result.

Another method is using vectors. Set O to be the origin. Then it is easy to show that $\bar{g} = \frac{\bar{a} + \bar{b} + \bar{c}}{3}$ and then use $(\bar{h} - \bar{a}) \cdot (\bar{b} - \bar{c}) = 0$. Here, it is easy to see that $h = \bar{a} + \bar{b} + \bar{c}$ works. Now the result is almost trivial.

[We derived the coordinates of these well known points again, since the popular method of originally finding them is using the Euler line. So, using the Euler line to prove the Euler line isn't fair, is it? Hence we gave an independent method from scrap.]

Lemma 1.14: $|OH|^2 = R^2(1 - 8\cos\alpha\cos\beta\cos\gamma)$

Proof : Note that the power of H with respect to $\odot(ABC)$ is $|OH|^2 - R^2$.

Now, we know by lemma 1.4 that $R_{BHA} = R$ and so by the law of sines applied to $\triangle ABH$, we get

$$AH = 2R\sin(90^\circ - \alpha) = 2R\cos\alpha$$

(This quantity is also important to know)

Also, by $\triangle BHD$, we get $|HD| = |BH|\sin(90^\circ - \gamma) = |BH|\cos\gamma = 2R\cos\beta\cos\gamma$.

Hence, Power of H with respect to $\odot(ABC)$ is $|AH| \cdot |HH_1| = 2|AH| \cdot |HD| = 8R^2\cos\alpha\cos\beta\cos\gamma$.

Hence, $|OH|^2 - R^2 = 8R^2\cos\alpha\cos\beta\cos\gamma$, as desired.

‡ A Special point

Here, we talk about the Miquel point of the (cyclic) quadrilateral $BFEC$. This point shows up many times in Olympiad problems, where it is usually defined by one of its properties listed below, but in fact, it comes free with many more! Try to recognize them!

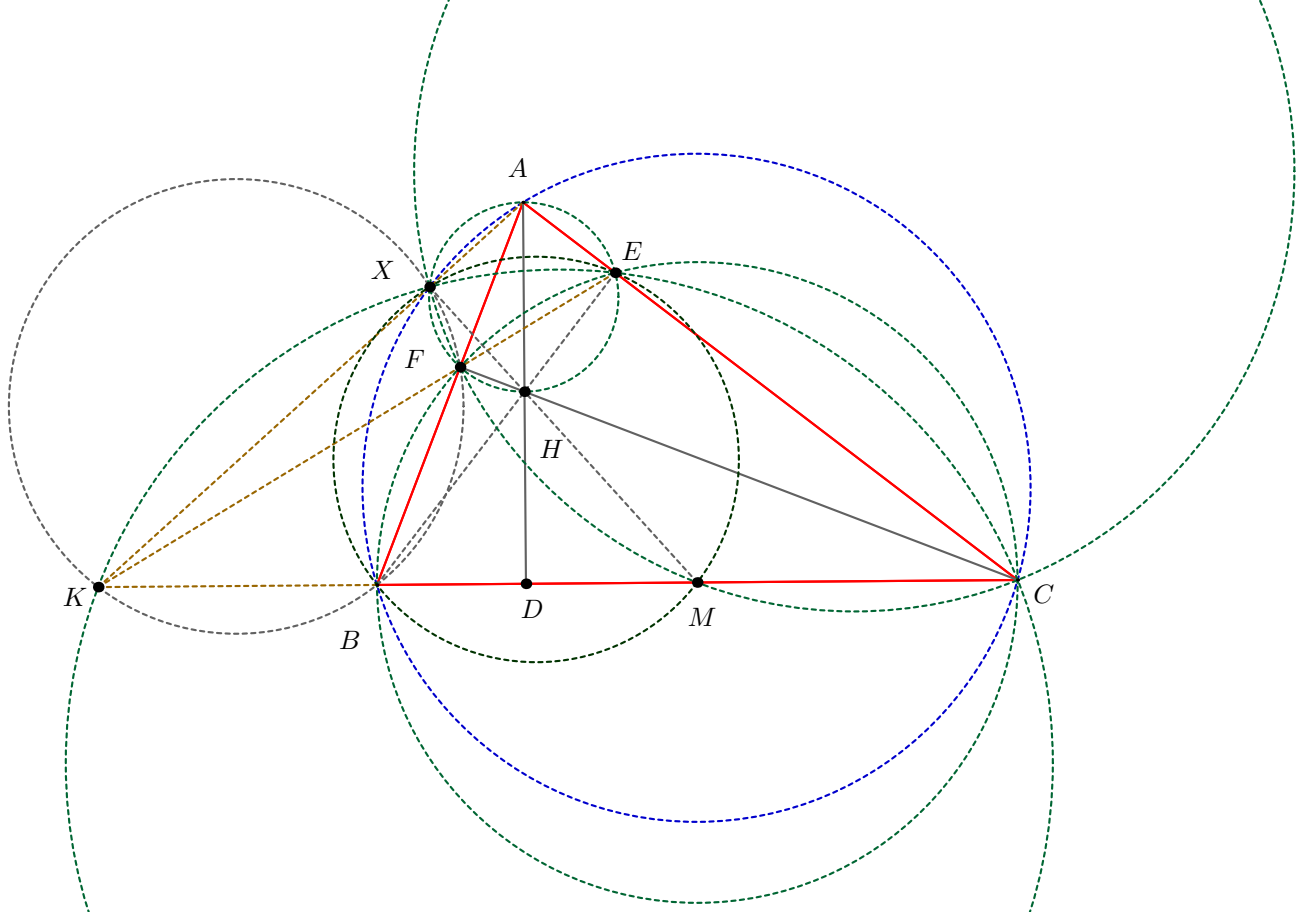


Figure 2: The Miquel Point X

Consider the point $X = MH \cap \odot(ABC)$ which is not the antipode of A . This special point possesses many properties:

All these are concurrent at X : $\odot(KBF), \odot(AEFH), \odot(ABC), \odot(MBE), \odot(KCE), \odot(FMC), AK, MH$. **Further, $MH \perp AK$.**

Many of these can be proved by the properties of the Miquel point, but we prove these independently.

Lemma 2.1 X lies on $MH, \odot(ABC), \odot(AEFH)$. **Further, $MH \perp AK$.**

Proof : Recall that X lies on MH and $\odot(ABC)$ by definition.

Now, by lemma 1.2, H_2 also lies on HM and is the antipode of A . Hence, $\angle HXA = \angle H_2XA = 90^\circ$ as AH_2 is a diameter. Thus, $MH \perp AK$. Also $\angle HXA = \angle HEA$ implying that X lies on $\odot(AEFH)$.

Lemma 2.2 K lies on AX .

Proof : Now, AX, EF and BC are the radical axes of $\odot(BFEC), \odot(AEFX)$ and $\odot(ABC)$, hence they are concurrent. But since $EF \cap BC = K$, hence K lies on AX .

Lemma 2.3 X lies on $\odot(KEC)$.

Proof : Now we show that $KXEC$ is cyclic. This follows since $\angle KEC = \angle FEC = 180^\circ - \beta$, as $BFEC$ is cyclic (by lemma 7) and

$\angle KXC = \angle KXM + \angle MXC = 90^\circ + \angle H_2XC = 90^\circ + \angle H_2AC = 90^\circ + \angle OAC = 180^\circ - \beta = \angle KEC$, and so X lies on $\odot(KEC)$.

Lemma 2.4 X lies on $\odot(FBK)$.

Proof : By using the result above and lemma 7, we get by Power of a Point $AX \cdot AK = AE \cdot AC = AF \cdot AB$, thus giving $XF BK$ is cyclic too.

Lemma 2.5 X lies on $\odot(BME)$.

Proof : By lemma 2.1, X lies on $\odot(ABC)$ and thus $\angle KXB = \gamma$. Also, by lemma 2.1 we know $\angle KXM = 90^\circ$ and thus $\angle BXM = 90^\circ - \gamma$.

Also, M is the center of $\odot(BEC)$ by lemma 1.7, and thus

$|ME| = |MB| \Rightarrow \angle BEM = 90^\circ - \gamma = \angle BXM$, implying that $XEMB$ is cyclic, as desired.

Lemma 2.6 X lies on $\odot(FMC)$.

Proof : By the lemma above, we know that $HX \cdot HM = HB \cdot HE$ by Power of a Point.

But also by lemma 1.7, we know that $BFEC$ is cyclic giving $HM \cdot HM = HB \cdot HE = HF \cdot HC$ and so $XFMC$ is cyclic, as desired.

Lemma 2.7 Let XD meet $\odot(ABC)$ again at P . Then AP is the A -symmedian.

Proof : Note that $AXDM$ is cyclic. Also, if A' is the antipode of A w.r.t $\odot(ABC)$, then A' lies on HM . Hence $\angle DAM = \angle DEM = \angle PAA'$ and so AP is a symmedian.

Now, we know that this point has many special properties, which will make our work easier... *if* this point is given in the question. But what if not?

Here is a way to bring this point out:

An inversion centered at A with radius $\sqrt{AH \cdot HD}$ sends $EF \cap BC$ to this point! And $EF \cap BC$ is in general, a commonly occurring point, and a useful construction (complete quadrilaterals anyone?)

Just as a side note, the point K itself is often a really useful construction, since one of its most important property (that gives it its sanctity) is the fact that $(KDBC)$ is harmonic, which often helps us find many lines that concur at K ! See, for instance, problem 21.

Another point (literally) worth noting is the image of M under $\mathbf{I}(A, \sqrt{AH \cdot AD})$. It is often called the HM -point and has a ton of useful properties. A document worth reading related to HM points is in the references [3]

‡ Simson and Steiner

The Simson Line is formed from a really interesting collinearity, and is closely related to the orthocenter by the Steiner line.

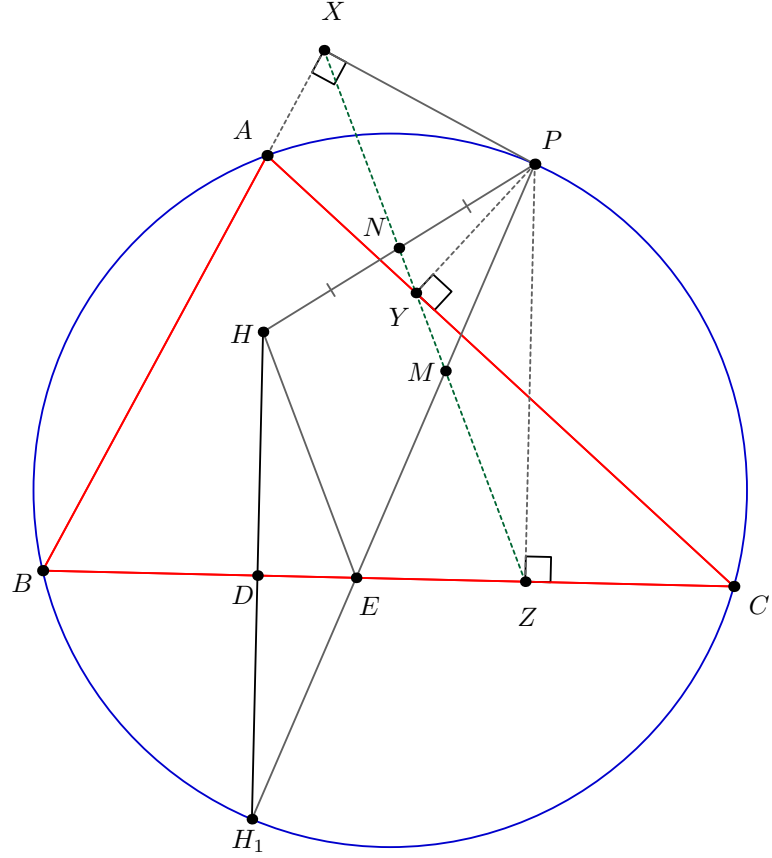


Figure 3: The P -Simson line of $\triangle ABC$

So the **Simson line** is defined as follows:

Consider any point P on $\odot(ABC)$. Then the feet of the perpendiculars from P to the respective sides of the triangle are collinear, and the line they determine is called the P -simson line of $\triangle ABC$

Proof : Note that the quadrilaterals $ABCP$, $PYZC$, $PXAY$ and $PXBZ$ are all cyclic. Thus,

$$\angle XYA = \angle XPA = 90^\circ + \angle XAP = 90^\circ + \angle BAP = 90^\circ + \angle BCP = 90^\circ + \angle ZCP = 90^\circ + \angle ZYP = \angle ZYC$$

giving the desired collinearity.

It is worth nothing that the converse of the above theorem is also true; i.e. These projections are collinear if and only if P lies on $\odot(ABC)$.

Now, time for the really important **Steiner line**:

Let R, S, T be the reflections of P over the sides BC, CA and AB respectively. Then R, S, T, H are collinear, where H is the orthocenter of $\triangle ABC$.

As some readers might have noticed, the Steiner line is obtained from the Simson line by a homothety centered at P with ratio 2. Then how is it any useful? The fact that the orthocenter lies on the Steiner line makes it, if not as useful, even more useful than the Simson line.

This leads to another form of the Steiner line; mentioned below

If H is the orthocenter of a $\triangle ABC$ and P is an point on $\odot(ABC)$, then the P -Simson line (which we call l) bisects PH .

Proof : This theorem is kind of hard to prove synthetically, but I present a beautiful proof following the idea from *Honsberger*:

Let H_1 represent the reflection of H over BC . By Lemma 1.1, we know that this point lies on $\odot(ABC)$. Let $PH_1 \cap BC = E$.

Now, since $PYZC$ is cyclic, hence $\angle YZE = \angle YPC = 90^\circ + \angle YCP = 90^\circ + \angle ACP = 90^\circ + \angle AH_1P = 90^\circ + \angle DH_1E = \angle DEH_1 = \angle HED$, where the last part follows since $\triangle HEH_1$ is isosceles.

Thus, we get that $HE \parallel l$.

Now, if M is the midpoint of PE , then $\triangle MZE$ is isosceles (since PZE is right angled) and so $\angle HED = \angle DEH_1 = \angle ZEM = \angle MZE$ and thus $MZ \parallel HE$. Thus, M lies on l .

Now since l is parallel to HE and M (which lies on l) is the midpoint of PE , hence we get that l is the P -midline of $\triangle PHE$ and so l bisects PH , as desired.

Remember, **Perpendiculars** and **reflections** over two sides are an infamous method of hiding the Simson and the Steiner line. Do not forget these two words!

‡ The Orthic triangle

The orthic triangle is the triangle formed by the feet of the altitudes.

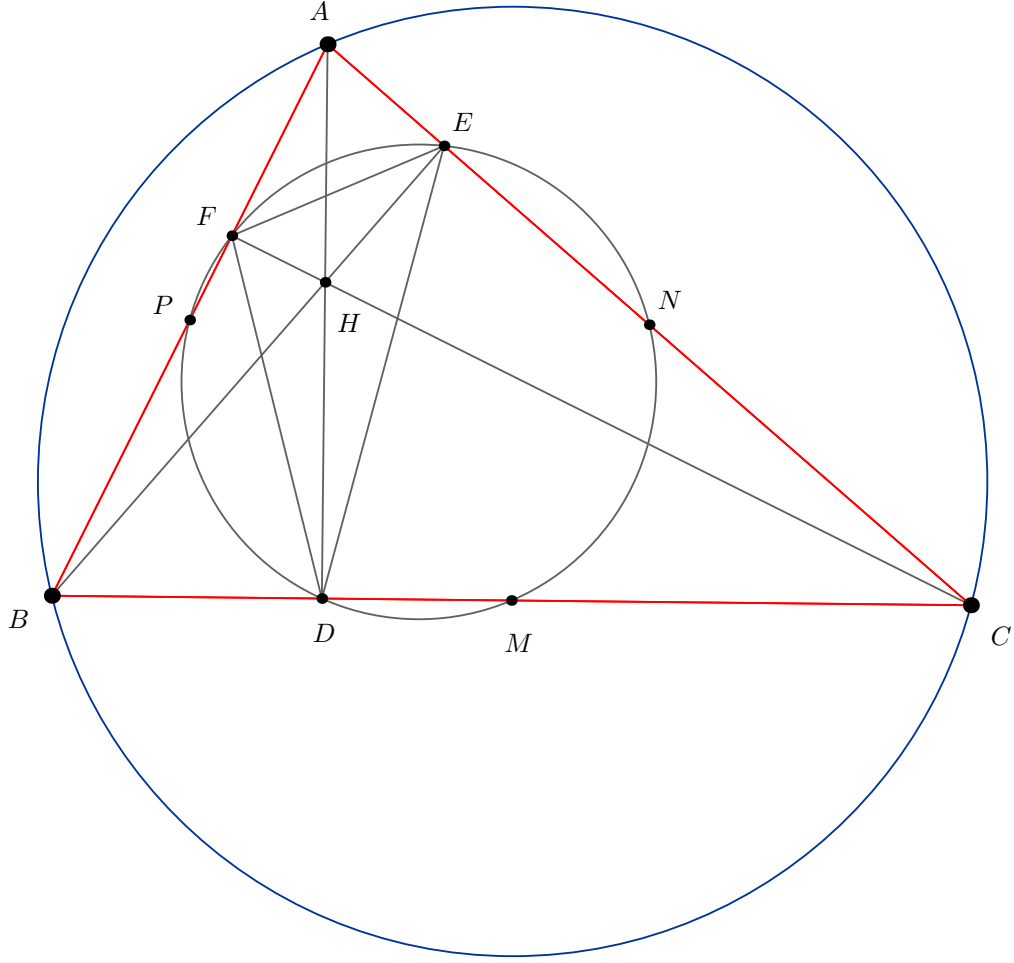


Figure 4: The Orthic Triangle

Though an obvious result from the Lemma 1.6, it is worth to keep in mind that

The circumcircle of the orthic triangle is the Nine-Point Circle, and hence passes through the midpoints of the sides of the main triangle.

Lemma 4.1 *H is the incenter of the orthic triangle.*

Proof : Note that since $FHDB$ is cyclic, hence $\angle HDF = \angle HBF = \angle EBA = 90^\circ + \angle CAB$ and $\angle EDH = \angle ECH = \angle ACF = 90^\circ + \angle CAB$ and so DH bisects angle FDE .

Similarly, we get that FH bisects $\angle DFE$ and so H is the the incenter of the orthic triangle.

The above result can also be directly prove by Blanchet's theorem, which we will soon talk about.

Lemma 4.2: *A, B, C are the D, E, F-excenters of the orthic triangle $\triangle DEF$, respectively*

Proof : Since $\odot(DEF)$ is the Nine-Point circle, hence it passes through X, Y, Z , i.e. the midpoints of AH, BH, CH respectively.

Now, by lemma 4.1, H is the incenter of $\triangle DEF$, and by a well known lemma, X is the midpoint of the incenter (i.e. H) and the D -excenter. Thus, the D -excenter is A . Similarly we get that B, C are the E, F excenters.

Fun fact: The Orthic triangle is the triangle with the minimum perimeter, whose vertices lie on the 3 sides of the triangle.

Proof : Since $\angle EDH = \angle HDF, \angle DFH = \angle HFE$ and $\angle FEH = \angle HED$, hence the orthic triangle is a ray of light, which thus covers the shortest path!

Another proof is by reflecting D over AB and AC , and considering the case when these reflections form a straight line with E, F , and then using lemma 4.2.

Lemma 4.3: (Sides of the orthic triangle) $|EF| = R\sin(2\alpha)$, $|FD| = R\sin(2\beta)$ and $|DE| = R\sin(2\gamma)$

Proof : By the sine rule on $\triangle AFE$, we get

$$|FE| = |AF| \cdot \frac{\sin(\alpha)}{\sin\beta} = b\cos\alpha \cdot \frac{\sin(\alpha)}{\sin\beta} = \cos\alpha\sin\alpha \cdot \frac{b}{\sin\beta} = \left(\frac{1}{2}\sin(2\alpha)\right)(2R) = R\sin(2\alpha)$$

and the other results follow similarly.

Hence, the perimeter of the orthic triangle (which is the minimum amongst other such triangles) is

$$R(\sin(2\alpha) + \sin(2\beta) + \sin(2\gamma)) = 4R\sin(\alpha)\sin(\beta)\sin(\gamma) = \frac{abc}{2R^2} = \frac{2[ABC]}{R}$$

where $[ABC]$ is the area of $\triangle ABC$.

‡ An Altitude Configuration

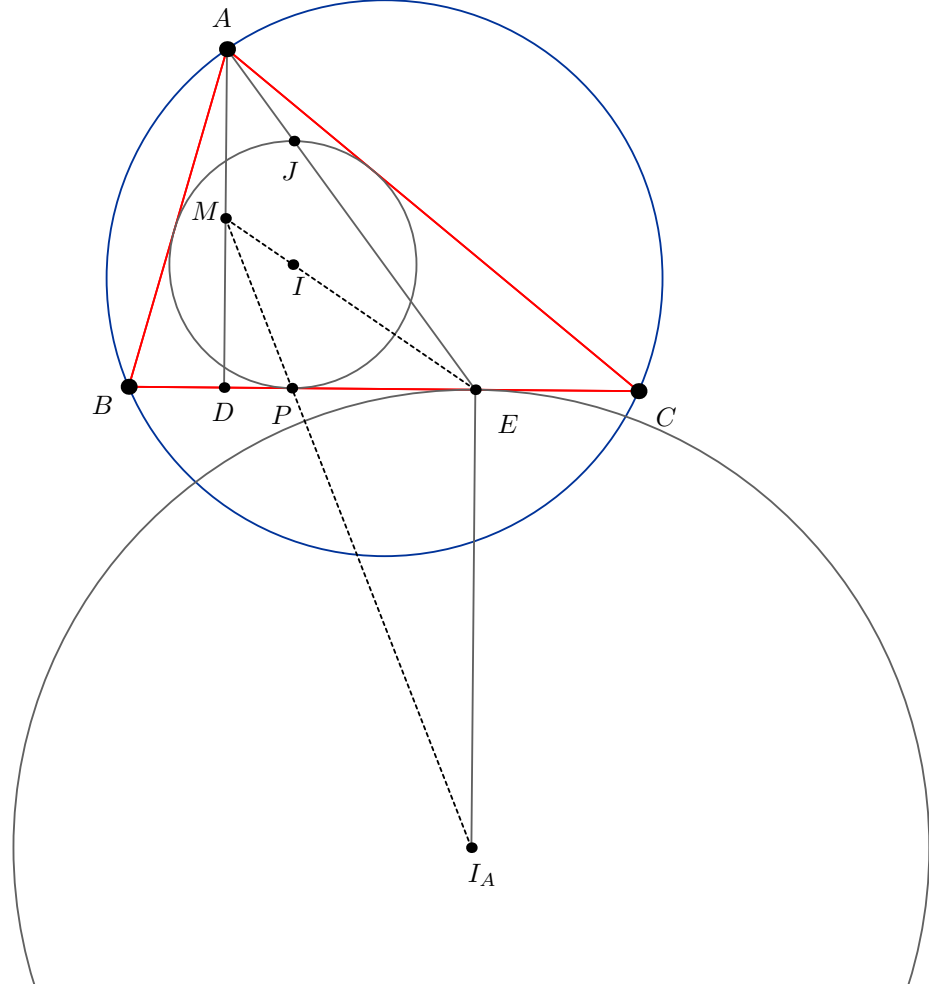


Figure 5: Midpoint of the Altitude

Here, define M to be the midpoint of the altitude AD , I to be the incenter of $\triangle ABC$, P, E to be the contact point of the incircle, A – excircle with BC , respectively. I_A is the center of the A -excircle.

Lemma 5.1 M, I and E are collinear.

Proof : Let J be the reflection of P over I . It is well known that this point lies on AE .

Now, since $JP \parallel AD$, we find a homothety centered at E taking $\triangle EJP$ to $\triangle EAD$, and so the midpoint of PJ , i.e. I , has the image M , i.e. the midpoint of AD . Hence, E, I and M are collinear, as desired.

Lemma 5.2: M, P and I_A are collinear.

Proof : Note that $AD \parallel I_A E$ and so there exists a homothety centered at P taking $\triangle PAD$ to $\triangle P Y E$, where Y is the antipode of E with respect to the A -excircle.

(Here we used the well known fact that A, P, Y are collinear. If you don't know this, then try a homothety centered at A taking the incircle to the A -excircle.)

Hence, the midpoint of the original altitude (M) goes to the midpoint of the new altitude (I_A). Hence, these points are collinear, as desired.

‡ Isogonal conjugates and Pedal triangles

Here, we include three useful properties of isogonal conjugates and pedal triangles that are generalizations of certain properties of orthocenters.

Theorem 6.1 Let P be a point in the plane of $\triangle ABC$ and let XYZ be the pedal triangle of ABC , i.e. X, Y, Z are the projections of P on BC, CA, AB respectively. Then the perpendiculars from A, B, C to YZ, ZX, XY respectively are concurrent at the isogonal conjugate of P with respect to ABC

Proof : This is just angle chasing and left to the reader.

Theorem 6.2 Let P be a point in the plane of $\triangle ABC$ and let D', E', F' be the reflections of P over BC, CA, AB . Then the isogonal conjugate Q of P with respect to ABC is the circumcenter of $D'E'F'$.

Proof : Consider a homothety centered at P with ratio $1/2$. This sends $D'E'F'$ to the pedal triangle of P , and so by theorem 6.1, AQ, BQ, CQ are perpendicular to the sides of $\triangle D'E'F'$.

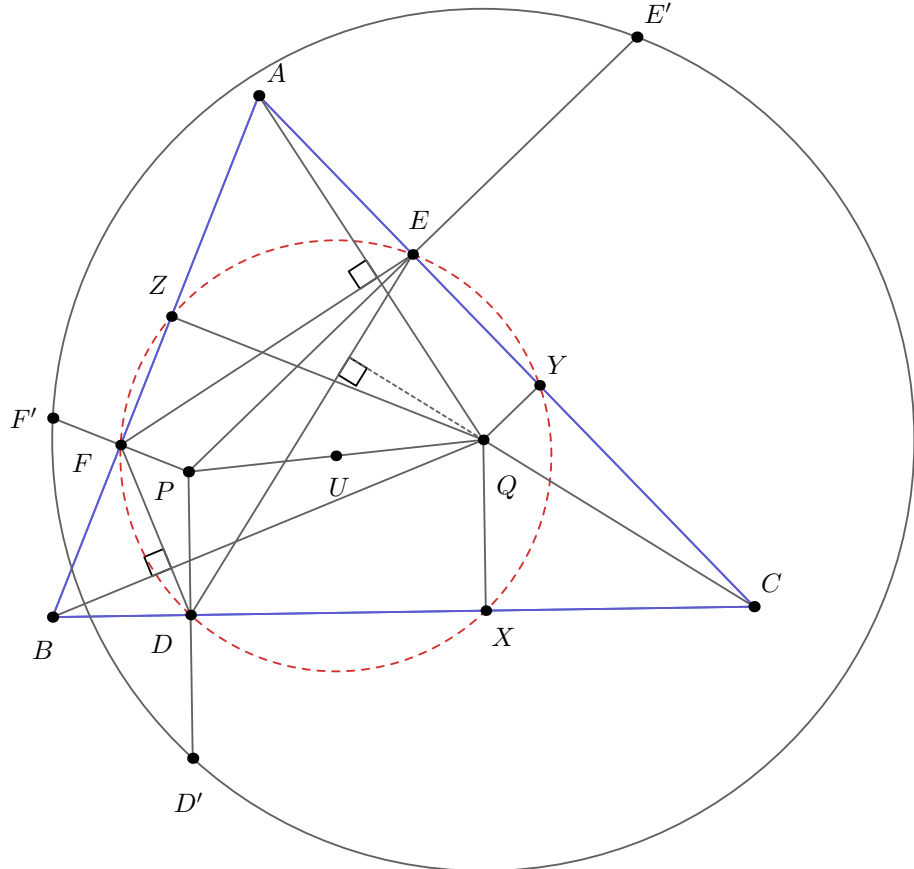
Now, $AF' = AP = AE'$ and $AQ \perp E'F'$. Hence, Q lies on the perpendicular bisector of $E'F'$. Similarly Q lies on the perpendicular bisectors of the other sides and so is the circumcenter of $\triangle D'E'F'$.

Now comes a beautiful theorem:

Theorem 6.3 (The Six Point Circle) Let P be a point in the plane of $\triangle ABC$ Q be the isogonal conjugate of P with respect to ABC . Let DEF, XYZ denote the pedal triangles of P, Q respectively with respect to ABC . Then D, E, F, X, Y, Z are concyclic point with respect to $\triangle ABC$. Further, the center of this circle is the midpoint of PQ .

Proof : Firstly, reflect P, Q about the sides BC, CA, AB and let the images be D', E', F', X', Y', Z' respectively. Then by theorem 6.2, we know that Q is the center of $\odot(D'E'F')$ and P is the center of $\odot(X'Y'Z')$. Hence, a homothety centered at P, Q with ratios $1/2$ yield that U , the midpoint of PQ , is the center of both $\odot(DEF)$ and $\odot(XYZ)$.

Now it suffices to show that these circles coincide and so it suffices to show $UD = UX$. But note that $PQXD$ is a trapezium with $PD \parallel QX$ and U is the midpoint of PQ . Hence U lies on the perpendicular bisector of DX . Hence U is equidistant from D, E, F, X, Y, Z and we are done!



Time for some application of these results to orthocenters. Firstly, note that the orthocenter(H) and the circumcenter(O) are isogonal conjugates (this is an extremely useful fact).

Try to prove these yourselves!

Corollary 1: Let DEF denote the orthic triangle of $\triangle ABC$. Then AO, BO, CO are perpendicular to the sides of $\triangle DEF$.

Corollary 2 (Bevan point): Let ABC be a triangle with A, B, C excenters I_a, I_b, I_c respectively. Further, let D, E, F be the tangency points of the A, B, C excircle with BC, CA, AB respectively. Then I_aD, I_bE, I_cF are concurrent at the circumcenter of $I_aI_bI_c$. This point is called the Bevan point of $\triangle ABC$.

Corollary 3: The reflections of H over BC, CA, AB lie on $\odot(ABC)$.

Corollary 4: Prove the existence of the nine-point circle.

Now that I have mentioned isogonal lines, I can't possibly forget to mention the amazing and incredibly useful *Isogonal Line Lemma*, which can be found in the references section. [6]

‡ Blanchet's theorem

Here is a marvelous result we present with 2 different, yet beautiful proofs! There is, of course, a trigonometric proof, but I decide to omit it.

Blanchet's theorem: Let ABC be a triangle with the altitude BY . Let P be any point on BY . Let $AP \cap BC = X$, $CP \cap AB = Z$. Then BY is the bisector of $\angle ZYX$.

Proof 1 : Construct a line l parallel to AC through B . Let $YZ \cap l = U$ and $YX \cap l = W$. Then it suffices to show $UB = WB$, as UYB will then be an isosceles triangle with the altitude YB and the result would follow.

Firstly, by Ceva's theorem,

$$\frac{AY}{YC} = \frac{XB}{CX} \cdot \frac{ZA}{BZ} \quad (1)$$

Now note that $\triangle UZB \sim \triangle YZA$ and $\triangle BWX \sim \triangle CYX$.

Hence,

$$\frac{XB}{CX} = \frac{WB}{YC}, \frac{ZA}{BZ} = \frac{AY}{UB} \quad (2)$$

Combining (1) and (2) yields the result.

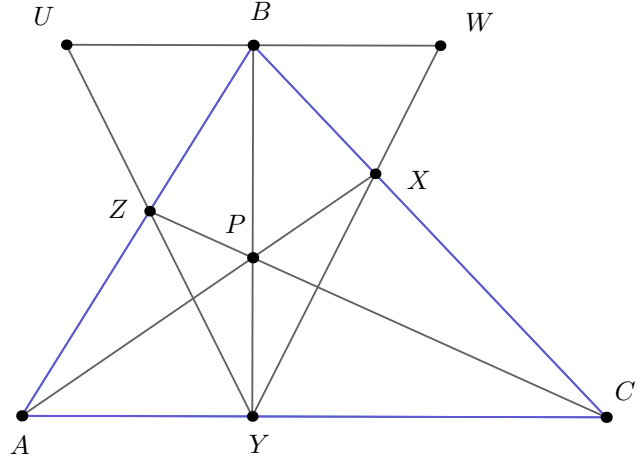


Figure 6: First proof

Proof 2 : Let $Y' = ZX \cap BY$ and $T = ZX \cap BC$. Then since AX, BY, CZ are concurrent cevians, hence (AC, YT) is harmonic. Projecting this onto line ZT (by a perspectivity at B), we get that $(ZX, Y'T)$ is harmonic. Now it is well known (for instance, via the Apollonius circle) that YY' bisects $\angle ZYX$, and so we are done.

Note that this proof gives us that as P varies over the altitude, the point T remains fixed.

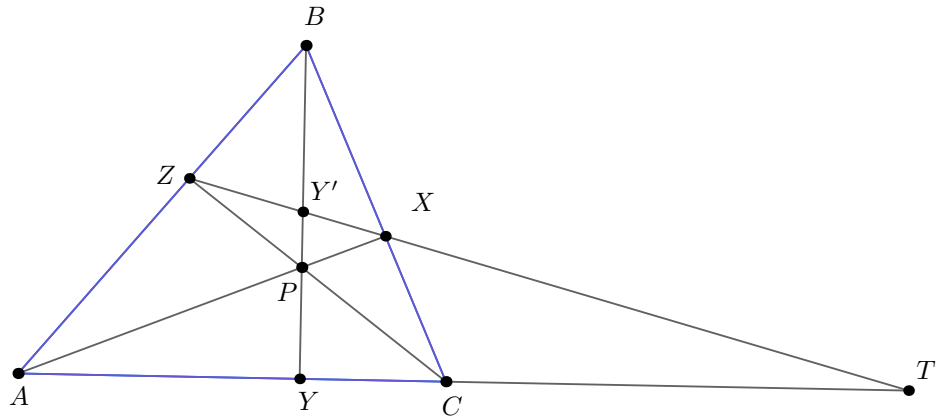


Figure 7: Second proof

We now present a nice generalisation of this result which appeared in the *2003 Indian National Mathematics Olympiad*.

Consider a triangle ABC and an interior point P . Let E, F be the intersection of BP, CP with AC, AB respectively. Let D be the intersection of AP and EF , and let K be the projection of D on BC . Then DK bisects $\angle FKE$.

Proof: The proof is the same, just extend EF and AP to meet BC . A nice non projective proof can be found in the references [8].

Here's yet another generalization. It is not so obvious as to how is this one a generalization. However, try to figure it out!

Let ABC be a triangle and P be a point in the plane of the triangle. Let DEF be the P -cevan triangle of ABC . Let M be the projection of D on EF . Then DM bisects $\angle BMC$.

Proof : Let $T = EF \cap BC$. Then clearly (TD, BC) is harmonic. Now since $\angle DMT = 90^\circ$, hence by a well known lemma DM bisects $\angle BMC$.

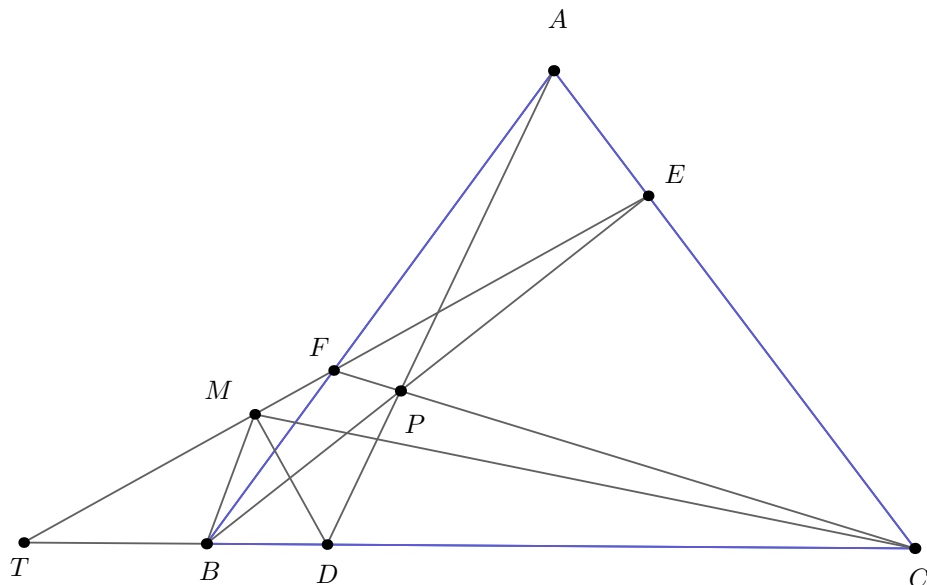


Figure 8: Generalization of Blanchet's theorem

We won't dwell much on Blanchet's theorem, but the interested reader can read up more about some beautiful applications of this in the book *Lemmas in Olympiad Geometry* given in the references. [2]

‡ Problem Pool

Here are some problems. Try to recognize the configurations and relate these to the lemmas. Some of these problems themselves might be useful lemmas!

Some of these problems might not use these lemmas and just be a standard problem, but these ones present some interesting and useful ideas, for instance for proving that $AP \perp BC$, you rather show that P is the orthocenter of $\triangle ABC$ etc.

Advice: As you might have noticed, complex numbers (or even vectors) are good friends of Orthocenters ;)

Lemma: Let ABC be an acute triangle and suppose Z is a point on $\odot(ABC)$ with $AZ \parallel BC$ and $Z \neq A$. Let G be the centroid of triangle ABC , and by D the foot of the altitude from A to BC . Prove that D, G, Z are collinear.

Problem 1 ([9] AoPS user Synthetic_Potato) Let ABC be an acute triangle. Let Ω_A be the circle with diameter BC . Similarly define Ω_B and Ω_C . Let AA_B be the tangent from A to Ω_A , closer to B . Similarly define points A_C, B_A, B_C, C_A, C_B . Then the points $A_B, A_C, B_A, B_C, C_A, C_B$ lie on a circle.

Problem 2 (USAMO 2010/1) Let $AXYZB$ be a convex pentagon inscribed in a semi-circle of diameter AB . Denote by P, Q, R, S the feet of the perpendiculars from Y onto lines AX, BX, AZ, BZ , respectively. Prove that the acute angle formed by lines PQ and RS is half the size of $\angle XOZ$, where O is the midpoint of segment AB .

Problem 3 (IMO 2013/4) Let ABC be an acute triangle with orthocenter H , and let W be a point on the side BC , lying strictly between B and C . The points M and N are the feet of the altitudes from B and C , respectively. Denote by ω_1 the circumcircle of BWN , and let X be the point on ω_1 such that WX is a diameter of ω_1 . Analogously, denote by ω_2 the circumcircle of triangle CWM , and let Y be the point such that WY is a diameter of ω_2 . Prove that X, Y and H are collinear.

Problem 4 (Hungarian Olympiad) Let $A, B, \dots, F$ be 6 points on a circle. $\triangle ABC$ and $\triangle DEF$ is considered a complementary pair of triangles. Consider the lines which connect the centroid of 1 triangle and the orthocenter of its complementary triangle. Prove that the 20 lines formed are all concurrent.

Problem 5 Let ABC be a triangle and let D, E, F be the feet of altitudes, with D on BC , E on CA and F on AB . Let the parallel through D to EF meet AB at X and AC at Y . Let T be the intersection of EF with BC . Prove that $\odot(TXY)$ passes through the midpoint of BC .

Problem 6 (USA TSTST 2012) In scalene triangle ABC , let the feet of the perpendiculars from A to BC , B to CA , C to AB be A_1, B_1, C_1 , respectively. Denote by A_2 the intersection of lines BC and B_1C_1 . Define B_2 and C_2 analogously. Let D, E, F be the respective midpoints of sides BC, CA, AB . Show that the perpendiculars from D to AA_2 , E to BB_2 and F to CC_2 are concurrent.

Problem 7 (IMO shortlist 1998 G3) Let I be the incenter of triangle ABC . Let K, L and M be the points of tangency of the incircle of ABC with AB, BC and CA , respectively. The line t passes through B and is parallel to KL . The lines MK and ML intersect t at the points R and S . Prove that $\angle RIS$ is acute.

Problem 8 Let $ABCD$ be a cyclic quadrilateral. The line through the midpoint of AB and perpendicular to CD is called an altitude of the quadrilateral. So, the quadrilateral has 4 conventional altitudes, and 2 unconventional altitudes (the diagonals). Prove that the 4 nine-point circles (of the 4 non-intersecting triangles formed by the diagonals), 4 Simson lines and the 6 altitudes are concurrent.

Problem 9 Let AD, BE, CF be the altitudes of a scalene triangle ABC with circumcenter O . Prove that $\odot(AOD), \odot(BOE)$, and $\odot(COF)$ intersect at a point X other than O .

Problem 10 (Iran TST 2011/1) In acute triangle ABC , $\angle B$ is greater than $\angle C$. Let M be the midpoint of BC and let E and F be the feet of the altitudes from B and C , respectively. Let K and L be the midpoints of ME and MF , respectively, and let T be on line KL such that $TA \parallel BC$. Prove that $TA = TM$.

Problem 11 (EGMO 2012/7) Let ABC be an acute-angled triangle with circumcircle Γ and orthocentre H . Let K be a point of Γ on the other side of BC from A . Let L be the reflection of K in the line AB , and let M be the reflection of K in the line BC . Let E be the second point of intersection of Γ with the circumcircle of triangle BLM . Show that the lines KH , EM and BC are concurrent.

Problem 12 (IMO shortlist 2000 G3) Let O be the circumcenter and H the orthocenter of an acute triangle ABC . Show that there exist points D , E , and F on sides BC , CA , and AB respectively such that

$$OD + DH = OE + EH = OF + FH$$

and the lines AD , BE , and CF are concurrent.

Problem 13 In a cyclic quadrilateral $ABCD$, points X and Y are the orthocenters of $\triangle ABC$ and $\triangle BCD$. Show that $AXYD$ is a parallelogram.

[I have found 3 solutions of this problem, 2 of which are based on lemmas from our list, and the third using... Try to find them (or even more)]

Problem 14 (BAMO 2013/3). Let H be the orthocenter of an acute triangle ABC . Consider the circumcenters of triangles ABH , BCH , and CAH . Prove that they are the vertices of a triangle that is congruent to ABC .

Problem 15 (USA TSTST 2011/4) Acute triangle ABC is inscribed in circle ω . Let H and O denote its orthocenter and circumcenter, respectively. Let M and N be the midpoints of sides AB and AC , respectively. Rays MH and NH meet ω at P and Q , respectively. Lines MN and PQ meet at R . Prove that $OA \perp RA$.

Problem 16 (Indian National MO 2017/4) Let ABC be a triangle with $\angle A = 90^\circ$ and $AB < AC$. Let AD be the altitude from A on to BC . Let P, Q and I denote respectively the incentres of triangles ABD , ACD and ABC . Prove that AI is perpendicular to PQ and $AI = PQ$.

Problem 17 (Hong Kong 1998) Let $PQRS$ be a cyclic quadrilateral with $\angle PSR = 90^\circ$ and let H and K be the feet of the altitudes from Q to lines PR and PS . Prove that HK bisects QS .

Problem 18 (USA TST 2014/1) Let ABC be an acute triangle, and let X be a variable interior point on the minor arc BC of its circumcircle. Let P and Q be the feet of the perpendiculars from X to lines CA and CB , respectively. Let R be the intersection of line PQ and the perpendicular from B to AC . Let ℓ be the line through P parallel to XR . Prove that as X varies along minor arc BC , the line ℓ always passes through a fixed point.

Problem 19 (USA TST 2011/1) In an acute scalene triangle ABC , points D, E, F lie on sides BC, CA, AB , respectively, such that $AD \perp BC, BE \perp CA, CF \perp AB$. Altitudes AD, BE, CF meet at orthocenter H . Points P and Q lie on segment EF such that $AP \perp EF$ and $HQ \perp EF$. Lines DP and QH intersect at point R . Compute HQ/HR .

Problem 20 (USA TSTST 2013/1) Let ABC be a triangle and D, E, F be the midpoints of arcs BC, CA, AB on the circumcircle. Line ℓ_a passes through the feet of the perpendiculars from A to DB and DC . Line m_a passes through the feet of the perpendiculars from D to AB and AC . Let A_1 denote the intersection of lines ℓ_a and m_a . Define points B_1 and C_1 similarly. Prove that triangle DEF and $A_1B_1C_1$ are similar to each other.

Problem 21 (USAJMO 2018/3) Let $ABCD$ be a quadrilateral inscribed in circle ω with $\overline{AC} \perp \overline{BD}$. Let E and F be the reflections of D over lines BA and BC , respectively, and let P be the intersection of lines BD and EF . Suppose that the circumcircle of $\triangle EPD$ meets ω at D and Q , and the circumcircle of $\triangle FPD$ meets ω at D and R . Show that $EQ = FR$.

Problem 22 (Sharygin 2016/12) Let BB_1 be the symmedian of a nonisosceles acute-angled triangle ABC . Ray BB_1 meets the circumcircle of ABC for the second time at point L . Let AH_A, BH_B, CH_C be the altitudes of triangle ABC . Ray BH_B meets the circumcircle of ABC for the second time at point T . Prove that H_A, H_C, T, L are concyclic.

Problem 23 (IMO Shortlist 1995/G3) The incircle of triangle $\triangle ABC$ touches the sides BC, CA, AB at D, E, F respectively. X is a point inside triangle of $\triangle ABC$ such that the incircle of triangle $\triangle XBC$ touches BC at D , and touches CX and XB at Y and Z respectively. Show that E, F, Z, Y are concyclic.

Problem 24 (2018 All Russian Olympiad Grade 10/P2) Let $\triangle ABC$ be an acute-angled triangle with $AB < AC$. Let M and N be the midpoints of AB and AC , respectively; let AD be

an altitude in this triangle. A point K is chosen on the segment MN so that $BK = CK$. The ray KD meets the circumcircle Ω of ABC at Q . Prove that C, N, K, Q are concyclic.

Problem 25 (ELMO 2018 P4) Let ABC be a scalene triangle with orthocenter H and circumcenter O . Let P be the midpoint of AH and let T be on line BC with $\angle TAO = 90^\circ$. Let X be the foot of the altitude from O onto line PT . Prove that the midpoint of PX lies on the nine-point circle of $\triangle ABC$.

Problem 26 (Sharygin 2012 Round 2 Grade 8) The altitudes AA_1 and CC_1 of an acute-angled triangle ABC meet at point H . Point Q is the reflection of the midpoint of AC in line AA_1 ; point P is the midpoint of segment A_1C_1 . Prove that $\angle QPH = 90^\circ$.

Problem 27 (ELMO 2018 Shortlist G4) Let $ABCDEF$ be a convex hexagon inscribed in a circle Ω such that triangles ACE and BDF have the same orthocenter. Suppose that BD and DF intersect CE at X and Y , respectively. Show that there is a point common to Ω , the circumcircle of $DX Y$, and the line through A perpendicular to CE .

Problem 28 (Sharygin 2011 round 2) Let BB_1 and CC_1 be the altitudes of acute-angled triangle ABC , and A_0 is the midpoint of BC . Lines A_0B_1 and A_0C_1 meet the line passing through A and parallel to BC in points P and Q . Prove that the incenter of triangle PA_0Q lies on the altitude of triangle ABC .

Problem 29 (ISL 2002, G7) The incircle Ω of the acute-angled triangle ABC is tangent to its side BC at a point K . Let AD be an altitude of triangle ABC , and let M be the midpoint of the segment AD . If N is the common point of the circle Ω and the line KM (distinct from K), then prove that the incircle Ω and the circumcircle of triangle BCN are tangent to each other at the point N .

Problem 30 (Iran TST 2009, Problem 9) In triangle ABC , D , E and F are the points of tangency of incircle with the center of I to BC , CA and AB respectively. Let M be the foot of the perpendicular from D to EF . P is on DM such that $DP = MP$. If H is the orthocenter of BIC , prove that PH bisects EF .

Problem 31 (Sharygin 2017 final round grade 9) Let BH_b, CH_c be altitudes of an acute-angled triangle ABC . The line H_bH_c meets the circumcircle of ABC at points X and Y . Points P and Q are the reflections of X and Y about AB and AC , respectively. Prove that $PQ \parallel BC$.

Problem 32 (China TST 2006) (Hagge Circle) Let ω be the circumcircle of $\triangle ABC$. P is an interior point of $\triangle ABC$. A_1, B_1, C_1 are the intersections of AP, BP, CP respectively and A_2, B_2, C_2 are the symmetrical points of A_1, B_1, C_1 with respect to the midpoints of side BC, CA, AB . Show that the circumcircle of $\triangle A_2B_2C_2$ passes through the orthocentre of $\triangle ABC$.

See [4] for more on the Hagge circle.

Problem 33 (ISL 2011 G4) Let ABC be an acute triangle with circumcircle Ω . Let B_0 be the midpoint of AC and let C_0 be the midpoint of AB . Let D be the foot of the altitude from A and let G be the centroid of the triangle ABC . Let ω be a circle through B_0 and C_0 that is tangent to the circle Ω at a point $X \neq A$. Prove that the points D, G and X are collinear.

Problem 34 (ISL 2015 G1) Let ABC be an acute triangle with orthocenter H . Let G be the point such that the quadrilateral $ABGH$ is a parallelogram. Let I be the point on the line GH such that AC bisects HI . Suppose that the line AC intersects the circumcircle of the triangle GCI at C and J . Prove that $IJ = AH$.

Problem 35 (APMO 2017 P1) Let H be the orthocenter of the triangle ABC . Let M and N be the midpoints of the sides AB and AC , respectively. Assume that H lies inside the quadrilateral $BMNC$ and that the circumcircles of triangles BMH and CNH are tangent to each other. The line through H parallel to BC intersects the circumcircles of the triangles BMH and CNH in the points K and L , respectively. Let F be the intersection point of MK and NL and let J be the incenter of triangle MHN . Prove that $FJ = FA$.

Problem 36 (USA TSTST 2016 P2) Let ABC be a scalene triangle with orthocenter H and circumcenter O . Denote by M, N the midpoints of $\overline{AH}, \overline{BC}$. Suppose the circle γ with diameter $\overline{AH}$ meets the circumcircle of ABC at $G \neq A$, and meets line AN at a point $Q \neq A$. The tangent to γ at G meets line OM at P . Show that the circumcircles of $\triangle GNQ$ and $\triangle MBC$ intersect at a point T on $\overline{PN}$.

‡ References and Further reading

1. Geometry Book:

Euclidean Geometry in Mathematical Olympiads by Evan Chen.

<http://web.evanchen.cc/geombook.html>

2. Geometry Book:

Lemmas in Olympiad Geometry by Titu Andreescu, Sam Korsky, and Cosmin Pohoata.

<https://www.maa.org/press/maa-reviews/lemmas-in-olympiad-geometry>

3. On *HM* points:

A Special Point on the Median by Anant Mudgal, Gunmay Handa.

https://www.awesomemath.org/wp-pdf-files/math-reflections/mr-2017-02/article_1_a_special_point_on_the_median.pdf

4. On the *Hagge* circle

Hagge circles revisited by Nguyen Van Linh.

<https://nguyenvanlinh.files.wordpress.com/2011/12/hagge-circles-revisited.pdf>

5. Midpoints of Altitudes blog:

Midpoints of Altitudes by Evan Chen.

https://artofproblemsolving.com/community/c1659h1031911_midpoints_of_altitudes

6. Isogonal line Lemma blog:

Isogonal Line Lemma by utkarshgupta.

https://artofproblemsolving.com/community/c2771h1181729_isogonal_line_lemma_updated

7. *Art of Problem Solving* website

<https://artofproblemsolving.com/>

Art of Problem Solving contest collection

https://artofproblemsolving.com/community/c13_contests

8. *2003 INMO* official solutions. (In particular, see problem 1 for the proof mentioned before.)

<http://olympiads.hbcse.tifr.res.in/olympiads/wp-content/uploads/2016/09/inmo-2003.pdf>

9. *Potato Circle* by Synthetic_Potato

<https://artofproblemsolving.com/community/q1h1640042p10329831>